CRYPTOGRAPHY AND NETWORK SECURITY

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1. CONCEPT MAP - 01 - SECURITY ATTACK, THREATS, RISKS AND SECURITY GOALS

INFORMATION TECHNOLOGY SECURITY

* Protecting data & systems from unauthorized access, changes, and disruption.

SECURITY ATTACKS

Interception : Unauthorized reading of data.
Modification : Unauthorized change
Interruption : Service is disrupted
Fabrication : False data is inserted.

SECURITY THREATS

Data exposure
Data tampering
Service disruption
Fake or unauthorized data

Attacks creates threats.

SECURITY RISKS

* Privacy violation • Financial loss • Incorrect records • Website or service downtimes.

SECURITY GOALS (CIA TRIAD) - mitigate the risks.

CONFIDENTIALITY

* Prevents unauthorized disclosure

Example : Encryption protects online banking data

INTEGRITY

* prevents or detects authorized changes.

Example : A digital signature helps verify a document.

AVAILABILITY

Keeps services accessible to authorized users.

Example : Backup services help a service continue during server failure.

Real world Links : A DDoS attack threatens availability and can cause a website outage. A stolen password threatness confidentiality. Unauthorized changes to bank records threaten integrity.

Conclusion

* Identify attacks, threats and risks helps organizations apply confidentiality, integrity and availablity control

CONCEPT MAP 2 : COMPARISON OF SUBSTITUTION TECHNIQUES

central concept : Substitution cipher replace plaintext symbols with other symbols.

SUBSTITUTION CIPHER

Key usage • complexity • Security strength

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1. CAESER CIPHER

Key : fixed shift number
Complexity : Very low
Security : Very weak ; only 25 non-zero shifts.
Example : CAT → FDW (shift +3)

2. Monoalphabetic cipher

Key : fixed mapping of each plaintext letter to a ciphertext letter
Complexity : low to use
Security : Weak against, but frequency analysis

3. VIGENERE CIPHER

Key : repeating keyword
Complexity : Higher than caeser
SECURITY : Stronger than simple substitution, but patterns can be analyzed.

4. HILL CIPHER

Key : Invertible matrix
Complexity : matrix based block operations
security : more complex, but vulnerable to known-plaintext attacks.

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COMPARISON

*Caesar : Simplest and weakest → Monoalphabetic : Larger key space but frequency pattern remain → Vigenere : multiple shifts using a keyword → Hill : Matrix-based block substitution.

ANALYSIS : WHICH IS MORE SECURE?

* There is no universal ranking between vigenere and hill without specifying the key and attack model. Both are stronger than caesar cipher and ordinary monoalphabetic substitution in many basic cases, but all four are unsuitable for modern sensitive data.

Conclusion :

* Increasing key complexity can improve resistance to simple attacks, but does not by itself guarantee security

3. CONCEPT MAP 3 : CLASSICAL ENCRYPTION.

Central concept : Classical encryption includes substitution and transposition.

CLASSICAL ENCRYPTION SCHEMES

convert plaintext into ciphertext using rules & key

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SUBSTITUTION TECHNIQES.

*Replace each letters with others
*character identify chances
*Basic example : Caesar cipher
Data structure : Order generally stays the same.

TRANSPOSITION TECHNIQUES

*Rearrange position of letters
*character identify the same
*Basic example : Rail fence cipher
*Data structure : Order changes.

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SECURITY WEAKNESS

*Simple substitution can be attacked using letter frequency patterns. Caesar has a very small key spaces.

SECURITY WEAKNESS

*Simple transposition preserves letter frequency ; an attacker can test possible arrangements and use language patterns.

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SIMILARITIES

Both hides the original message, use a defined rule/key, produce ciphertext and can be reversed by decryption. In basic forms, both usually preserve message length.

ANALYSIS : WHICH IS MORE VULNERABLE ?

* Simple caesar substitution is espicially vulnerable because because its key space is tiny. Simple transposition is also vulnerable because it preserves letter frequencies. Vulnerability depends on the specific cipher, key space and attack method

CONCLUSION :-

* Substitution changes symbols identifies, while transposition changes symbol position. Neither simple form provides modern security.

4. CONCEPT MAP 4 : STEGANOGRAPHY - DATA HIDING, WATERMARKING AND SURVIVABILITY

Central concept : Steganography conceals information inside an ordinary digital medium

STEGANOGRAPHY

Hiding the existence of a message in an image, audio, video or other file

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1. DATA HIDING

* Purpose : Conceal that secret information exists
* Relationship : A message is embedded inside a cover file.
* Application : Covert communication between authorized parties

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2. WATERMARKING

* Purpose : Indicate ownership or support authenticity checks.
* Relationship : An identifying mark is embed in digital content
* Application : Copyright identification for digital photographs

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3. SURVIVABILITY (ROBUSTNESS)

* Purpose : Help an embedded message remain detectable after expected changes.
* Relationship : Robust embedding aims to withstand operation such as compression or resizing.
* Application : Detecting a copyright watermark after image compression.

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HOW THEY SUPPORT SECURE COMMUNICATION

* Data hiding conceals the presence of information, watermarking supports ownership / authenticity, survivability improves resistance to expected modification.

Conclusion :

* Those concepts serve different purposes and can complement each other, but steganography alone does not guarantee confidentiality, encryption may also be needed.